EXERCISE 2

1. SELECT DISTINCT department
FROM students;

Output:

department
IT
HR
Finance

2. SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department;

Output:

department	avg_age
IT	20.5
HR	22
Finance	23

3. SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;

Output:

department	student_count
IT	2
HR	2

4. SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;

Output:

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;

Output:

department	total_credits
IT	11

7. SELECT course_id, course_name, department, credits
FROM courses
WHERE credits <> 4 ;

Output:

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3 ;

Output:

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. SELECT
MAX (grade) AS max_grade,
MIN (grade) AS min_grade,
AVG (grade) AS avg_grade
FROM enrollments ;

Output:

max_grade	min_grade	avg_grade
90	78	84.6

10. SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;

Output:

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. SELECT department
SUM(salary) AS total_salary
SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;

Output:

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;

Output:

department	avg_salary
IT	81000
Finance	70000

13. SELECT employee_id, name, salary, bonus,
(salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60000 ;

Output:

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14. SELECT department,
SUM (budget) AS total_budget,
AVG (budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000 ;

Output:

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

15. SELECT project_id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';

Output:

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000